FINOPTIX

SUMMER PROJECT '25

Black Litterman Model

INTRODUCTION

- The **Black-Litterman model** is an advanced portfolio allocation method developed by Fischer Black and Robert Litterman at Goldman Sachs. It combines modern portfolio theory (mean-variance optimization) with subjective investor views, while grounding everything in market equilibrium.

- **Core Idea**: The Black-Litterman model is a portfolio optimization framework that combines market equilibrium (implied returns from market-cap weights) with an investor's personal views on asset performance. It enhances traditional mean-variance optimization by producing more stable and diversified portfolios. The model adjusts expected returns based on the confidence level in each view, avoiding extreme allocations. It reduces sensitivity to estimation errors in historical returns and blends data-driven insights with expert judgment. Widely used in institutional investing, Black-Litterman offers a balanced, flexible, and practical approach to portfolio design.

READING RESOURCES

- Black Litterman Model (Reading Resource 1)
- Black Litterman Model (Reading Resource 2)

PYTHON IMPLEMENTATIONS

- Black Litterman Model (Python Implementation)
- MVO Optimization (Python Implementation)
- Fama French Factors Dataset Link

Assignment-4 Link